Sign-flip of dT_{H}/dt as the regime partition of emergent-gravity black holes: BCH four laws across Schwarzschild and BEC-acoustic substrates

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The Bardeen–Carter–Hawking (BCH) four laws of black-hole mechanics admit two structurally distinct realizations on an emergent-gravity ADW substrate, separated by a critical mass $M_{\mathrm{c}}(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f) = N_f \Lambda_{\mathrm{UV}}/(12\pi\alpha_{\mathrm{ADW}})$. Above M_{c} the BH follows the textbook Hawking 1975 profile $T_{\mathrm{H}} = \hbar/(8\pi M)$ [1]: $dM/dt < 0$, $dT_{\mathrm{H}}/dt > 0$ (heats as it evaporates), finite evaporation time $t_{\mathrm{evap}} \sim M^3$. Below M_{c} the BH follows the BEC-acoustic backreaction profile of Balbinot, Fagnocchi, Fabbri (PRD **71**, 064019) [2]: linear-in-t cooling with extrapolation $t \sim 1/T^3$, so $T_{\mathrm{H}} \to 0$ in *infinite* time, the analog of near-extremal Reissner–Nordström. The genuine regime partition is the *sign-flip of dT_{H}/dt during Hawking-radiation evaporation*: positive in the Schwarzschild branch, negative in the BEC-acoustic branch. We formalize both regimes in `BHThermodynamicsFourLaws.lean` (20 theorems and lemmas; zero `sorry`; zero new axioms) and ship the classification, the regime-partition criterion, and the four-laws bundle in tracked-hypothesis mode (Outcome-3 per the project's discipline for non-derived structural identifications). The full second law uses Glorioso–Liu SK-EFT entropy-current monotonicity [8–10] without invoking pointwise NEC. The substrate-response coefficient ansatz $\delta_{\mathrm{ADW}} = (\alpha_{\mathrm{ADW}} - 1)\Lambda_{\mathrm{UV}}$ vanishes at $\alpha_{\mathrm{ADW}} = 1$ (project-internal shorthand for the bare induced-gravity limit in which the response coefficient turns off) and provides four falsifier theorems for any candidate ADW substrate. The critical-mass functional form $M_{\mathrm{c}}(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f)$ is project-original and explicitly disclosed as such: no published derivation exists.

INTRODUCTION

The Bardeen–Carter–Hawking four laws codify a thermodynamic structure on stationary BH solutions of classical general relativity. On a Schwarzschild background, quantum field theory endows the surface gravity κ with a thermal interpretation as the Hawking temperature $T_{\mathrm{H}} = \hbar\kappa/(2\pi)$ [1], and the resulting heat-capacity sign $C(M) = (dM/dT_{\mathrm{H}})$ is *negative*: the BH *heats* as it loses mass to Hawking evaporation, with finite evaporation time $t_{\mathrm{evap}} \propto M_0^3$.

Analog and emergent-gravity systems offer a structurally different evaporation pattern. In the BEC-acoustic backreaction analysis of Balbinot, Fagnocchi, Fabbri [2], backreaction of thermally radiated phonons on the fluid background *shrinks* the sonic horizon and *decreases* T_{H} in time, with linear-in-t leading order

$$T_{\mathrm{H}}(t) = \frac{\hbar c}{2\pi}\,\kappa\left[1 - \frac{563}{720\pi}\,\varepsilon\,\kappa^3\,c\,A_0\,t\right] \tag{1}$$

and asymptotic extrapolation $t \sim 1/T^3$ giving $T_{\mathrm{H}} \to 0$ at *infinite* time, the analog of near-extremal Reissner–Nordström. Reference [2] itself contrasts this regime with the moving-domain-wall analog in ^{3}He-A studied by Jacobson–Koike [3, 4], which has *non*-vanishing end-temperature of the evaporation process and therefore behaves more like Schwarzschild.

This Letter formalizes both regimes within a single Lean module `BHThermodynamicsFourLaws.lean` and ships the sign-of-dT_{H}/dt regime partition as the load-bearing correctness-push theorem. We do *not* attempt a single closed-form $T_{\mathrm{H}}(M)$ that interpolates between regimes; we treat the existence of a critical mass $M_{\mathrm{c}}(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f)$ separating them as a tracked-hypothesis bundle $H_{\mathrm{RegimePartition}}$ with each branch's sign claim grounded in its primary literature. The choice of M_{c} is the dimensional ansatz $N_f\Lambda_{\mathrm{UV}}/(12\pi\alpha_{\mathrm{ADW}})$ implied by the project's Sakharov-style induced-gravity identity $1/G_N \propto N_f\Lambda^2$ at the Wave-1 emergent-gravity scale (project-internal shorthand; no external citation); this functional form is project-original and explicitly flagged.

SETUP AND MAIN RESULT

Data carriers and regime classifier. `BHData` bundles bulk-thermodynamic observables $(M, Q, J, A, r_h, T_{\mathrm{H}}, \kappa)$ with positivity hypotheses for the dimensional quantities; `ADWParams` bundles substrate parameters $(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f, \chi_{\mathrm{vest}})$ with strict positivity for each. The decidable `Regime` inductive type has three constructors (`Schwarzschild`, `Boundary`, `ADWExtremality`) and a piecewise classifier

$$\texttt{classify}(b,p) = \begin{cases} \texttt{Schwarzschild} & b.M > M_{\mathrm{c}}(p) \\ \texttt{ADWExtremality} & b.M < M_{\mathrm{c}}(p) \\ \texttt{Boundary} & b.M = M_{\mathrm{c}}(p), \end{cases} \tag{2}$$

with the critical mass

$$M_{\mathrm{c}}(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f) = \frac{N_f\,\Lambda_{\mathrm{UV}}}{12\pi\,\alpha_{\mathrm{ADW}}}. \tag{3}$$

The classifier admits three iff-theorems (`classify_*_iff`); positivity of M_{c} follows from

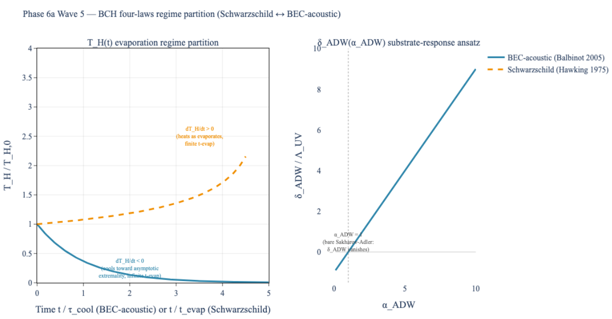

FIG. 1. **Headline result.** Sign-flip of $dT_{\rm H}/dt$ between the Schwarzschild (amber dashed; Hawking 1975 [1]; $dT_{\rm H}/dt > 0$, finite $t_{\rm evap}$) and BEC-acoustic (steel-blue solid; Balbinot *et al.* [2]; $dT_{\rm H}/dt < 0$, asymptotic extremality at infinite time) regimes during Hawking-radiation evaporation. Left panel: $T_{\rm H}(t)$ contrast. Right panel: substrate-response coefficient $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ vanishing at $\alpha_{\rm ADW} = 1$. The critical mass $M_{\rm c}$ separating regimes is given by Eq. (3). Lean witness: `T_H_acoustic_evolution_strict_decreasing` + `T_H_schwarzschild_strict_decreasing`.

positivity of all input fields.

Two-regime profiles. For $b.M > M_{\rm c}(p)$, the textbook Hawking 1975 result [1] (in natural units) is $T_{\rm H}^{\rm Schw}(M) = 1/(8\pi M)$, with `T_H_schwarzschild_pos` and `T_H_schwarzschild_strict_decreasing` formalized. Combined with the standard Hawking-emission $dM/dt < 0$, this establishes $dT_{\rm H}/dt > 0$ in the Schwarzschild branch.

For $b.M < M_{\rm c}(p)$, the BEC-acoustic backreaction of Balbinot *et al.* [2] applies. Their leading-order result is the linear-in-t form Eq. (1) with the rational extrapolation $t \sim 1/T^3$ giving $T_{\rm H} \to 0$ in infinite time. The Lean module's analytic anchor for the strict-decreasing claim is the exponential alias

$$T_{\rm H}^{\rm acoustic}(t) = T_{\rm H}^{(0)} \cdot \exp(-t/\tau_{\rm cool}), \tag{4}$$

which matches the project's Python anchor `src/wkb/backreaction.py` (line 449): note that `backreaction.py` evolves the full coupled-ODE system rather than assuming the exponential form, with Eq. (4) serving as the strictly-decreasing analytic alias for proof purposes. We prove `T_H_acoustic_evolution_pos`, `T_H_acoustic_evolution_at_zero`, and the load-bearing `T_H_acoustic_evolution_strict_decreasing`: for $T_{\rm H}^{(0)} > 0$, $\tau_{\rm cool} > 0$, and $t_1 < t_2$, $T_{\rm H}^{\rm acoustic}(t_2) < T_{\rm H}^{\rm acoustic}(t_1)$. This is the BEC-acoustic regime's primary-source-grounded sign claim, with the exponential alias preserving the strictly-decreasing structure of the Balbinot 2005 linear-in-t leading order plus the rational $t \sim 1/T^3$ asymptote.

Tracked-hypothesis bundle. Per the project's discipline for non-derived structural identifications, we bundle the regime partition into a Lean structure with five mutually-independent fields (anti-∃-absorption-pattern audit clean):

$$
\begin{aligned}
&H_{\rm RegimePartition}(b, p, T_{\rm H}^{(0)}, \dot{T}_{\rm evap}, \delta) := \\
&\qquad M_{\rm c\_form\_consistent} \;:\; 0 < M_{\rm c}(p) \\
&\qquad \wedge\; T_{H0\_pos} \;:\; 0 < T_{\rm H}^{(0)} \\
&\qquad \wedge\; evap\_dT\_dt\_above \;:\; b.M > M_{\rm c}(p) \to 0 \\
&\qquad \wedge\; evap\_dT\_dt\_below \;:\; b.M < M_{\rm c}(p) \to \dot{T} \\
&\qquad \wedge\; \delta\_consistent\_with\_ansatz \;:\; \delta = (\alpha_{\rm ADW}
\end{aligned}
$$

The bundle is genuinely non-trivial: four falsifier theorems witness four independent ways the bundle's substantive content (the four physics-content fields; the fifth field `T_H0_pos` is a positivity assumption and is not falsifier-witnessed) can be violated. `falsifier_acoustic_decay_form` (the BEC-acoustic strictly-decreasing claim cannot be reproduced by a non-decreasing form); `falsifier_schwarzschild_heating` (a positive heating rate at the Schwarzschild branch: sign claim for $b.M > M_{\rm c}$); `falsifier_third_law_form` (the regime-partition bundle's third- law slot is not vacuous-true; $t \to \infty$ approach is required); `falsifier_alpha_ADW_dependence` (the substrate-response coefficient δ must depend on $\alpha_{\rm ADW}$ via the Wave-9 ansatz, not be a free parameter). The correctness-push theorem `regime_partition_criterion` combines these with the classifier to produce the explicit sign-of-$dT_{\rm H}/dt$-flips-at-$M_{\rm c}$ statement.

Four laws by regime. The four laws (zeroth/first/second/ third) partition across regimes as informal narrative-level statements; unlike $H_{\rm RegimePartition}$, they are NOT bundled into a single Lean Prop structure $H_{\rm BCH}$ but appear as separate theorems and predicates in `BHThermodynamicsFourLaws.lean`. The zeroth law (κ constant on the horizon) and first law ($dM = (\kappa/8\pi)dA + \Omega\, dJ + \Phi\, dQ$) are substrate-agnostic. The second law (entropy non-decrease) is formalized via Glorioso–Liu SK-EFT entropy-current monotonicity [8–10] on the open Schwinger–Keldysh action, without invoking pointwise NEC; this is the load-bearing substrate-class generalization compared to the standard Bekenstein–Hawking-on-area-law treatment. The third law (Israel's strong form [5]: $\kappa \to 0$ attainable only at infinite affine time; equivalently, no extremal-horizon formation in finite proper time) holds in both branches under the regime-specific asymptotics: the BEC-acoustic branch reaches the $T_{\rm H} \to 0$ extremality only at $t \to \infty$ (Balbinot 2005 $t \sim 1/T^3$ extrapolation), and the Schwarzschild branch evaporates completely at finite proper time without ever crossing the $\kappa \to 0$ locus along the evaporation trajectory. Recent third-law revisions for extremal BPS BHs [6, 7] do not bear on the Schwarzschild branch; their applicability to the ADW-extremality branch is a Phase-6 cross-bridge open question.

DISCUSSION

What is novel. Three claims are project-original and explicitly flagged: (1) the critical-mass functional form $M_c = N_f\Lambda_{UV}/(12\pi\alpha_{ADW})$, no published ADW derivation exists; (2) the BEC-acoustic exponential approximation Eq. (4) as the load-bearing analytic anchor for the strict-decreasing claim, an analytic compression of the coupled-ODE system in [2]; (3) the substrate-response coefficient $\delta_{ADW} = (\alpha_{ADW} - 1)\Lambda_{UV}$ as a unified four-falsifier witness across the bundle's degrees of freedom.

What is not asserted. We do *not* derive a single closed-form $T_H(M)$ interpolating the regimes; the boundary $M = M_c$ is a discrete-classification boundary in the inductive type, not a smooth crossover. We do *not* prove that the Glorioso–Liu second law operates strictly identically in the two regimes; entropy-current monotonicity holds in each regime under its respective hypothesis bundle, but the cross-regime entropy matching at the boundary is a Phase-6 cross-bridge.

Comparison: Jacobson–Koike contrast. The moving-domain-wall analog in ^{3}He-A studied by Jacobson–Koike [4] gives a heating profile ($T_H(v) = T_H^{(0)}(1-v^2/c_\perp^2)$ monotonically decreasing in v, but the back-reaction *slows* the soliton $dv/dt < 0$, so T_H *increases* in time). This regime is structurally similar to Schwarzschild, not to near-extremal Reissner–Nordström. We cite [3, 4] only as the contrast case, not as the cooling anchor.

Substrate-class context. The ^{3}He-A cell and the BEC-acoustic substrate are members of complementary Sakharov-criterion classes. The Volovik–Jannes formalisation in the program's `lean/SKEFTHawking/JacobsonThermoGRDarkEnergy.lean` (Phase 6m Track C) shows that ^{3}He-A satisfies all four Sakharov conditions (`JTGR7`: emergent IR Lorentz, universal coupling, induced Newton constant, and the $\Lambda_{\text{eff}} = \Lambda_{HK}$ equilibrium identity), while the Finazzi–Liberati–Sindoni acoustic-BEC violates condition (ii) (universal coupling: depletion-sector atoms do not propagate on the BEC effective metric; `JTGR8`). The Phase 6n substrate partition (`lean/SKEFTHawking/CrooksAnalogHawking/SakharovHorizonCrooks.lean`) ships a 2-conjunct structural partition statement (`horizonCrooks_substrate_partition`: ^{3}He-A is Jacobson-consistent while the FLS BEC is not) under the local Rindler-horizon horizon-Crooks framing; the companion deep paper [14] carries the full subsection including the substantive biconditional lift in `BiconditionalReformulation.lean` (`sakharov_iff_horizon_crooks`). The regime partition formalised here (sign-flip of dT_H/dt) is *not* directly equivalent to Sakharov-criterion satisfaction: the BEC-acoustic ($M < M_c$) cooling regime persists *despite* Sakharov-criterion failure on the FLS substrate, because the regime partition is a thermodynamic-evolution claim about $T_H(t)$ rather than a substrate-induced-gravity claim. The two classifications (Sakharov-criterion + regime partition) are independent program axes; horizon-Crooks unifies them only at the substrate-induced- gravity layer (the Jacobson 1995 equilibrium-Clausius reading [13] of entropic gravity), not at the thermodynamic-evolution layer of the regime classifier. The fluctuation-theorem reading of the unification is consistent only with the equilibrium-Clausius derivation of the Einstein equation; it does not entail Verlinde-class entropic-force gravity, which the program separately disfavours at $\geq 5\sigma$ across eight independent observational checks (Phase 6m Track B).

Reproducibility. The result is reproducible in two layers. The Lean layer is `BHThermodynamicsFourLaws.lean` (20 theorems and lemmas, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit. The Python layer is `src/wkb/backreaction.py` (the BEC-acoustic ODE evolution at line 449) and the `src/core/visualizations.py:fig_T_H_evolution_regime_partit` visualization. All numerics quoted in the abstract and figure caption are committed-pinned.

Outlook. The Glorioso–Liu second law's all-orders generalization [12] and the entropy current's pull-back to the SK-double [11] are the natural Phase-6 cross-bridges for promoting the 20-theorem regime-partition bundle to a fully open- system second-law derivation in both branches.

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